

2. → Quantum Systems: Electrons, Photons and Atoms.

→ Quantum Systems are microscopic systems that follow the law of quantum mechanics. The most important quantum systems are.

- Electrons
- Photons
- Atoms - Understanding these concepts/systems helps us understand modern technologies, such as computers, mobile phones, lasers and communication systems.

(a) → Electrons: An electron is a negatively charged subatomic particle present in every atom. Electrons exist around the nucleus of an atom and are responsible for electrical and electronic phenomena.

Real-time Analogy: - Consider a ceiling fan rotating at a very high speed.

- When the fan rotates slowly, individual blades are visible.
- When it rotates very fast, the blades appear as a blurred circular region.

Electrons are responsible for,

- Electricity generation and transmission
- operation of computers
- Mobile phones
- Internet devices
- Electronic circuits. Without electrons, modern electronic technology would not exist.

(b) → Photons:

→ A photon is the fundamental particle of light. Photons carry electromagnetic energy and travel at the speed of light.

Real-time
Example.

Q. What reaches your eyes when you look at a mobile phone screen?

A. Photons emitted from the screen reach your eyes, allowing you to see images and text.

Application of Photons.

Photons are used in

- mobile phones display
- Television screens
- Lasers
- Fibre-optic Communication
- Optical Sensors.

Fact → A laser light pointer ~~emits~~ emits billions of photons every second. Quantum communication systems can use individual photons to securely transmit information.

→ Photons are the basic carrier of light and information in optical communication systems.

(c) → Atoms

→ An atom is the smallest unit of an element that retains its chemical properties.

→ Atoms consist of:

- Nucleus (protons and neutrons)
- Electrons surrounding the nucleus.

Real-Life

Example

→ consider a piece of chalk. If it is broken into smaller and smaller pieces repeatedly, eventually the smallest units obtained are atoms.

→ Importance of Atoms:

- Atoms are the building blocks of
 - Human beings
 - Water
 - Air
 - Buildings
 - Earth
 - All matters in the universe, Everything around us is ultimately made up of atoms.

QUANTIZATION AND DISCRETE ENERGY LEVELS

→ Quantization is one of the fundamental concepts of quantum mechanics. It states that certain physical quantities, such as energy, can take only specific discrete values rather than any arbitrary value.

Example

→ Stair case Analogy.

→ Consider a stair case.

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You stand on

→ step 1 } But you cannot stand between two steps.
→ step 2 } Similarly, electrons in an atom can occupy
→ step 3 } only specific energy levels. They cannot exist
at intermediate energy values.

Elevator
Analogy → Imagine a building with multiple floors. you can stay on

floor 1
floor 2
floor 3, But not on floor 2.5. Likewise, electron
can exist only in allowed energy levels. When energy is
absorbed or released, electrons moves from one energy
levels to another.

Electron Transitions.

Energy Absorption: When an electron absorbs energy
it jumps from a lower energy levels to a higher energy
levels. This process is called excitation.

Energy Emission: When an electron returns to a
lower energy level.

→ It releases energy.

→ The released energy appears as light (photon).

Real-Time Example - Real-life Examples of Quantization

Neon Sign Boards: Electrons gain energy and move
to higher energy levels. When they return to lower
levels, they emit colored light, producing the glow seen
in neon signs.

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LED lights: LEDs work because electrons release photons when moving between energy levels.

Fire works: Different chemicals contains atoms with different energy levels. When electrons return to lower energy states, different colors of light are produced.

Therefore:

- Electrons are negatively charged particles that exist as probability clouds around the nucleus.
- Photons are particles of light and carriers of electromagnetic energy.
- Atoms are the fundamental building blocks of matter.
- Quantization means energy exists in discrete levels.
- Electrons can occupy only specific energy states.
- Movement of electrons between energy levels results in absorption or emission of light.
- Application of quantization include LEDs, lasers, neon sign and fireworks.

→ Thus, Electrons, photons, and atoms are the fundamental quantum systems that form the basis of modern technology. The concept of quantization explains why energy exists in discrete levels and help us understand many real-world phenomena such as light emission, LEDs, and electronic devices.

SNAPSHOT OF QUANTUM TECHNOLOGIES & GLOBAL QUANTUM MISSIONS.

The Quantum Technologies are,

- (i) Quantum Computing
- (ii) Quantum Communication
- (iii) Quantum Sensing

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1. → Quantum Computing: Quantum computing is a new computing technology that uses qubits instead of classical bits. By using quantum properties such as superposition and entanglement, quantum computers can solve certain complex problems much faster than classical computers.

Example: Imagine searching for a specific book in a library containing 10 million books.

- classical computer: ~~search~~ searches one shelf after another
- Quantum computer: can explore many possibilities simultaneously and find the solution more efficiently for specific problems.

Application of Quantum Computing.

① Drug Discovery

- Scientists spend years of developing new medicines
- Quantum computers can simulate molecular interaction more accurately.
- Faster discovery of drugs of interest.

② Weather and Climate prediction

- Weather forecasting requires enormous calculation
- Quantum computers can improve climate and weather models.

③ Financial Optimization

- Banks and financial institutions use optimization for
 - * Investment planning
 - * Risk reduction
 - * Portfolio management
 - * Quantum Computing can help find optimal solutions faster.

Quantum Computing aims to solve complex problems beyond the practical capabilities of classical computers.

② → Quantum Communication

→ Quantum Communication uses the principles of quantum mechanics to provide highly secure communication.

→ In classical communication, hackers may intercept information without being detected. Quantum communication enables the detection of eavesdropping attempts.

Example: Imagine sending a sealed envelop containing a secret key.

Applications:

- Banking transactions
- Military communication
- Government Network
- Healthcare records
- Cyber security.

Quantum Internet: The quantum internet is a future communication network, where quantum computers are interconnected. Information is transmitted using quantum states. Communication becomes highly secured.

→ Quantum Communication provides secure information transfer by exploiting the laws of quantum mechanics.

③ → **Quantum Sensing:** Quantum sensing uses quantum phenomena to achieve extremely precise measurements.

- A sensor is a device used to measure physical quantities such as
- Temperature
 - Pressure
 - Motion
 - Magnetic fields.

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→ Quantum sensors provide much higher accuracy than conventional sensors.

Applications:

- GPS Improvement Improves navigation accuracy
Reduces positioning error
- Medical Imaging Enhances MRI and diagnostic scans
Helps detect diseases at earlier stage
- Earth Exploration Detection of
 - oil reserve
 - underground water
 - Minerals

Comparison of Quantum Technologies

Tech	Main Purpose	Example Application
Quantum Computing	Fast problem solving	Drug discovery
Quantum Communication	Secure communication	Quantum Internet
Quantum Sensing	Precise measurement	GPS and medical image process

National and Global Quantum Missions.

→ Quantum technology is expected to revolutionize computing, communication, and sensing, similar to how electronic computers and the internet transformed society.

Major goals include:

- National security
- Economic growth
- Technological leadership
- Scientific advancement
- Future employment opportunities